

# **HUMAN-ASSISTED HYBRID INTERACTION CONTROL TOWARD ROBOTIC 3D ULTRASOUND ACQUISITION**

**Regular MEng Project Final Report**

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## **Statement on the Use of Generative Artificial Intelligence**

Generative artificial intelligence was used as an assistive tool during the preparation of this report. OpenAI ChatGPT and Codex were used to help reorganize draft sections, identify repetition and internal inconsistencies, improve grammar and clarity, and suggest explanations suitable for readers without a robotics background. The tools were also used to support document formatting and the preparation of original explanatory diagrams.

Generative AI also assisted with reviewing the controller source code and preparing scripts for descriptive analysis and visualization of the recorded CSV logs. These uses were limited to technical checking and presentation support. All AI-assisted text, code suggestions, calculations, and figures were examined against the controller implementation, source logs, and project records, then revised or rejected by the author as needed.

The project direction, controller objectives, implementation decisions, parameter tuning, safety procedures, physical operation of the Franka Research 3 robot, and experimental data collection were undertaken by the author with supervisory and laboratory support. The author made the final decisions about which trials were reported and how the results and limitations were interpreted. Generative AI was not used to operate the robot, fabricate measurements, or create experimental results. No patient or clinical data were used in this project.

References proposed during AI-assisted work were checked against the original articles, publisher records, or official technical documentation before inclusion. The arguments, conclusions, and recommendations in this report remain the author's responsibility. This disclosure is intended to distinguish editorial and analytical assistance from the engineering work and experimental evidence on which the report is based.

## Abstract

Three-dimensional ultrasound combines a sequence of two-dimensional images acquired from known probe poses into a volumetric representation. Unlike an individual 2D image, the reconstructed volume can be reviewed from multiple planes and can support spatial and volumetric measurements after acquisition. In future autonomous robotic acquisition, the robot must move the probe across the patient while maintaining continuous acoustic contact and regulating the probe-skin contact force around a suitable target. This requires coordinated probe motion, contact-force measurement, pose tracking, spatial calibration, and synchronized ultrasound acquisition.

This project developed and tested a human-guided controller for the Franka Research 3 as an early step towards autonomous robotic 3D ultrasound. The operator could reposition the end effector by hand, while the controller adjusted its vertical equilibrium using a contact-force estimate derived from joint torque. The final program also included software limits, controlled stopping, and timestamped data logging.

In selected stationary-contact periods, the controller kept its internal force estimate close to the target. The estimate became unreliable during manual guidance because it could not separate the operator's input from surface contact and other robot-related effects. For this reason, the current system is best viewed as an interaction and logging prototype, not an autonomous or force-validated ultrasound platform. The next stages require direct force sensing and probe integration, followed by synchronized image and pose acquisition, spatial calibration, and phantom testing.

**Keywords:** Franka Research 3; robotic ultrasound; 3D ultrasound; Cartesian impedance; physical human–robot interaction; contact-force estimation.

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## Nomenclature

$q, \dot{q}$ : Joint-position and joint-velocity vectors

$J(q)$ : Geometric Jacobian

$M(q), c(q, \dot{q}), g(q)$ : Inertia matrix, Coriolis/centrifugal term, and gravity term

$\tau, \tau_{ext}$ : Commanded and external joint-torque vectors

$\dot{x}$ : Cartesian end-effector twist

$F_{ext}$ : External Cartesian wrench

$K, D$ : Virtual stiffness and damping coefficients

$OT_{EE}$ : End-effector pose relative to the robot base

$\widehat{OF}_{ext,K}$ : FR3 model-based external wrench estimate associated with stiffness frame K

$\widehat{F}_{contact}$ : Rectified torque-derived contact estimate,  $\max(-F_{z,corr}, 0)$

$v_{des,xy}$ : Bounded desired XY velocity generated by the guidance logic

$z_d$ : Desired Z equilibrium position

$Q, Q_d$ : Measured and desired unit quaternions

$R$ : Measured end-effector rotation matrix

$e_R$ : Orientation-error vector

$W_{task}$ : Commanded Cartesian task wrench

$e_F$ : Estimated contact-force error

$v_{cmd}$ : Fixed XY command-speed magnitude

**FR3**: Franka Research 3

**TCP**: Tool centre point

**RMSE**: Root-mean-square deviation from the nominal internal target

# 1. Introduction

## 1.1 Background and Motivation

Ultrasound imaging uses a transducer, commonly called a probe, to transmit high-frequency sound waves into the body and receive returning echoes. A conventional B-mode image converts the echo strength into a two-dimensional greyscale cross-section. The method is non-ionizing, portable, comparatively inexpensive, and capable of displaying anatomy in real time, which makes it useful for many diagnostic and image-guided procedures.

The image depends on how the clinician holds and moves the probe. Probe position, orientation, contact pressure, scan speed, and coupling all affect the acquired image. The clinician continuously adjusts the probe to preserve acoustic contact while avoiding unnecessary pressure on the patient. Freehand ultrasound is flexible, but repeated scans may not follow the same path or apply the same pressure [1]. These variations can change image appearance and reduce the geometric consistency needed when several images are combined or compared.

For a future autonomous system, these adjustments cannot depend on continuous clinician input. The robot must maintain sufficient and approximately constant probe-skin contact as it moves across the patient. This is necessary to preserve acoustic coupling while limiting discomfort, excessive tissue deformation, and variation between acquired images.

## 1.2 From 2D Ultrasound to a 3D Volume

A three-dimensional ultrasound volume can be constructed from a sequence of conventional 2D B-mode images. During a sweep, each image must be paired with the position and orientation of the probe at the time of acquisition. This combined description is the probe pose. Spatial calibration relates pixels in the ultrasound image to the physical tool and tracking frames. The registered slices can then be transformed into a common coordinate system and compounded into a volumetric representation [2].

Figure 1.1 summarizes the three required stages. Each B-mode frame must share a timestamp with the measured robot pose. Spatial calibration supplies the fixed relationship from the robot tool to the ultrasound image plane, allowing each slice to be placed in a common robot

or world frame before volume compounding. Timing, pose, calibration, loss of acoustic contact, and pressure-induced tissue deformation can all reduce reconstruction consistency [2].

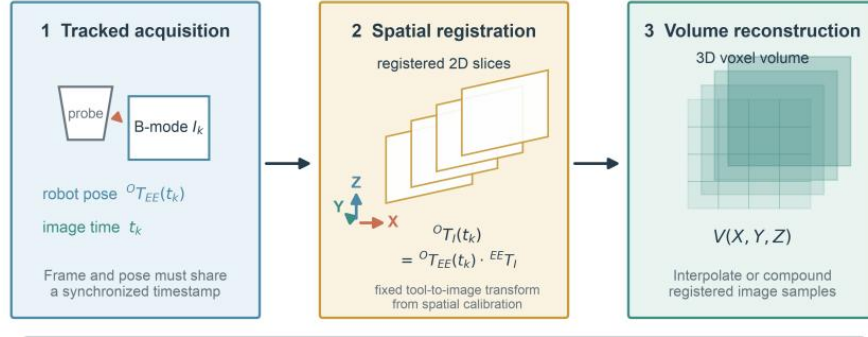


Figure 1.1. Tracked 2D ultrasound acquisition, spatial registration, and volume reconstruction.

A reconstructed 3D ultrasound volume can be viewed from different planes after the scan and used for spatial or volumetric measurements. It also records the scanned region as a volume instead of showing only the single image plane selected by the operator at a given moment. This is useful for anatomical visualization, comparison between examinations, and image-guided procedures.

In the planned system, the FR3 would provide tracking information through its measured end-effector pose. After calibrating the transformation between the end effector and the ultrasound image plane, each 2D image could be placed in the correct position for reconstruction. In this report, “robotic 3D ultrasound” therefore means that the robot supports a conventional 2D probe and records its pose. The robot itself does not produce a 3D ultrasound image.

### 1.3 Human-Guided Robotic Ultrasound

The long-term goal is autonomous robotic 3D ultrasound acquisition. This project focuses on an earlier, human-guided stage. In the intended system, a trained operator would choose the scan path and interpret the live ultrasound image, while the robot would assist with probe motion and contact regulation. This keeps the operator in control without yet requiring full surface perception, autonomous path planning, or image-based decision making.

The Franka Research 3 (FR3) was selected because its seven-degree-of-freedom, torque-controlled architecture supports Cartesian impedance control and physical interaction. In this

report, compliance refers to the robot’s ability to yield to an externally applied force rather than rigidly resisting displacement. The degree of compliance is determined mainly by the selected Cartesian stiffness and damping. Lower stiffness allows the operator or the environment to displace the end effector more easily, while damping limits rapid motion and oscillation.

Each Cartesian direction has a main role, but none is assigned exclusively to either the operator or the robot. The operator mainly guides the scan in the horizontal XY plane and can also move the end effector in Z because the vertical axis remains compliant. At the same time, the controller adjusts the desired Z equilibrium from the estimated contact-force error. Low rotational stiffness lets the operator adjust the orientation by hand, and the end effector starts in a downward-facing pose for scans on a horizontal surface.

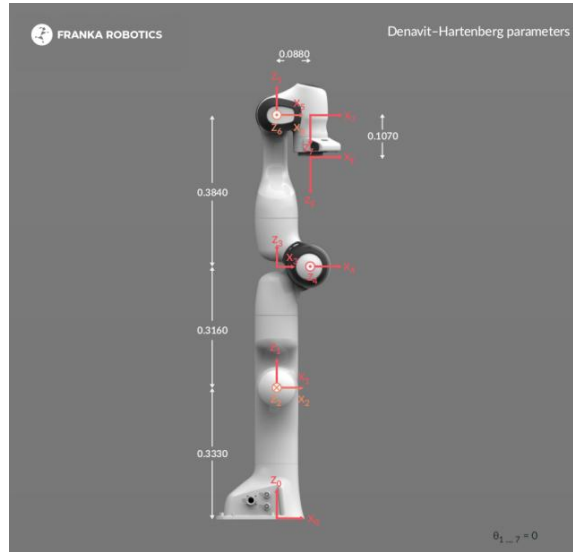


Figure 1.2. Official kinematic coordinate-frame definition of the Franka Research 3 in its zero-joint configuration. Frame 0 defines the robot-base origin. Reproduced from Franka Robotics [8].

The origin of the robot-base coordinate frame is fixed at the base of the FR3, as shown by frame 0 in Figure 1.2. The controller reports the end-effector pose relative to this base frame. The positive Z axis extends upward from the base, while X and Y span the base plane. For the experimental arrangement used in this project, the robot-base XY plane was approximately parallel to the horizontal test surface, and Z was treated as the surface-normal direction. This description applies to the reported laboratory setup and does not define the future ultrasound image coordinate system.

The project combines Cartesian impedance control, robot dynamics, external-wrench estimation, safety limits, and data logging for a future robotic ultrasound system. The main questions are whether the operator can reposition the robot with low resistance and whether the available force estimate is suitable for an ultrasound task that requires continuous contact.

## 1.4 Project Aim and Contribution

The project aimed to develop and evaluate a Franka-based human-guided control framework for future robotic 3D ultrasound acquisition. The immediate objective was to build a compliant controller that permitted manual translation, used XY motion mainly to define the scan path, retained manual Z adjustment, superimposed contact-related Z-equilibrium adaptation, allowed tool-orientation adjustment, and recorded the variables needed for evaluation.

The main implementation contribution is the progression from v1 to v5.1 within a standalone libfranka Cartesian impedance architecture. The final prototype combines manually guided XY motion, Z-equilibrium adaptation based on the FR3 torque-derived wrench, and compliant orientation control. The FR3 has seven rotating joints numbered from the robot base towards the end effector. Joint 6 is the second-to-last joint in this chain and is located in the wrist region near the end effector. Reducing the controller-generated torque contribution at this joint made the tool easier to rotate manually. The prototype also includes software limits, controlled termination, and buffered CSV logging.

The main limitation turned out to be force sensing, not path generation. Without a direct end-effector sensor, the FR3 wrench estimate combines probe contact, operator input, configuration-dependent effects, and modelling error. The completed work is therefore an interaction and logging prototype. Probe mounting, direct force sensing, synchronized ultrasound acquisition, spatial calibration, and 3D reconstruction were not completed in this project.

## 1.5 Report Structure

Chapter 2 reviews robotic ultrasound and cooperative control literature. Chapter 3 describes the development route, final system concept, objectives, and project boundaries. Chapter 4 presents the theoretical framework, Chapter 5 documents the controller implementation and experimental procedure, and Chapter 6 analyses the logged results. Chapter 7 summarizes the conclusions, recommendations, limitations, and future work.

## 2. Background and Literature Review

### 2.1 Existing Research in Robotic 3D Ultrasound

Chapter 1 introduced the acquisition chain for robotic 3D ultrasound. Existing research has examined several components of this chain separately, including robot-assisted probe motion, pose tracking, spatial calibration, volume reconstruction, and contact regulation. Janvier et al. evaluated the geometric performance of a robotic 3D ultrasound imaging system [22], while Krupa investigated automatic calibration between the robot-mounted probe and the ultrasound image plane [23]. These studies show that accurate robot motion alone is not sufficient. Reconstruction also depends on synchronized image and pose measurements and a calibrated transformation between the robot and ultrasound image coordinate systems.

A robot can move the probe more systematically than an unconstrained freehand sweep. Autonomous scanning would also require the robot to follow the body surface, maintain a suitable probe-skin force, and respond to changes in image quality or patient motion [4]-[6]. The reconstruction problem is therefore closely tied to the contact-control problem studied in this report.

A reconstructed 3D ultrasound volume can be used for visualization, spatial measurements, and automated anatomical analysis. In the example in Figure 2.1, Zhang et al. process a fetal 3D ultrasound volume to detect landmarks, locate standard planes, and measure crown-rump length [3]. This method was not implemented here; it is included as an example of how a consistently acquired volume could later be used.

This project addresses an earlier part of the pipeline: manually repositioning the FR3, providing contact-related assistance, and recording the variables needed to evaluate the controller. Ultrasound acquisition, probe-to-image calibration, and volume reconstruction were outside the completed work.

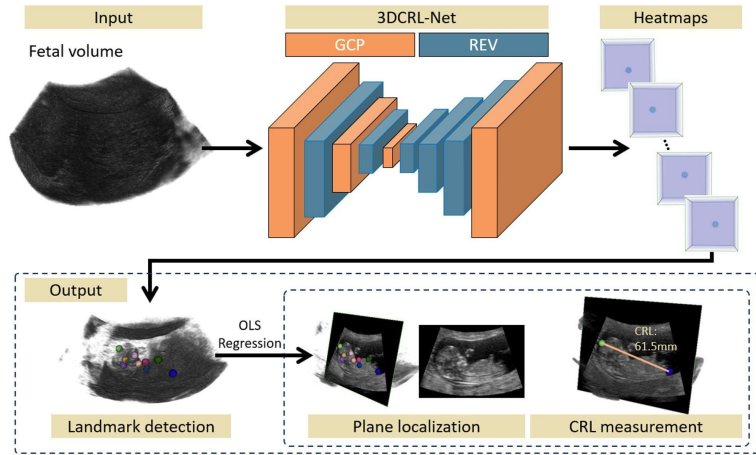


Figure 2.1. 3DCRL-Net workflow for fetal landmark detection, plane localization, and CRL measurement. Reproduced without modification from Zhang et al. [3] (CC BY-NC-ND 4.0).

## 2.2 Robotic Ultrasound Control Approaches

Robotic ultrasound systems are usually described as autonomous, teleoperated, or cooperative [1]. In practice, these categories can overlap. A system may, for example, let a person choose the path while the robot regulates contact force. Early work combined teleoperation with force and image feedback, and later studies used commercial collaborative robots for compliant ultrasound control [19]–[21]. This project falls mainly in the cooperative category.

Table 2.1. Principal robotic ultrasound control approaches.

| Mode         | Human role                                     | Main benefit                                              | Main challenge                                                  |
|--------------|------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------|
| Autonomous   | Selects the target and supervises the scan.    | Repeatable path execution and systematic coverage.        | Requires perception, planning, registration, and robust safety. |
| Teleoperated | Controls the probe through a remote interface. | Extends access to specialist expertise.                   | Introduces latency, interface, and force-feedback issues.       |
| Cooperative  | Physically guides the robot-held probe.        | Preserves intuitive manipulation while the robot assists. | Must distinguish guidance input from contact force.             |

Repeatable robot motion alone is not enough for autonomous ultrasound. A clinician normally coordinates surface following, acoustic contact, probe pressure, image coverage, and responses to patient motion at the same time. An autonomous robot would need to handle these



tasks safely [5], [6], which makes reliable force sensing and contact regulation necessary parts of the system.

The current controller does not provide these autonomous functions. The operator still selects the path, while the robot allows manual repositioning and adjusts the Z equilibrium in response to the estimated contact force. This setup makes it possible to study interaction and sensing before adding autonomous path planning, surface following, or image-based decisions.

## 2.3 Cooperative and Hybrid Control

Cooperative control assigns complementary roles to the human and robot. The human can select the anatomical region and respond to unexpected imaging conditions, while the robot can support the probe, apply motion limits, and repeat movements. This allocation is consistent with the broader physical human-robot collaboration literature, in which robot repeatability is combined with human adaptability under explicit interaction and safety constraints [25]. For ultrasound, useful interaction requires more than low resistance in every direction: the probe must remain easy to reposition while preserving a controlled contact state [19]-[21].

Impedance control creates a virtual spring-damper relationship by generating forces or torques from position and velocity errors [14]. Admittance control instead maps measured force to commanded motion and is widely used in physical human-robot interaction [17]. The present controller combines related elements: joint-torque commands produce Cartesian impedance and damping, while the XY guidance logic maps the direction of the estimated external force to a bounded desired velocity.

The Cartesian directions are not divided into separate human-controlled and robot-controlled axes. The operator mainly guides the scan in XY but can also move the robot manually in Z through the compliant impedance. Meanwhile, the controller changes the desired Z equilibrium according to the contact-force error. The resulting Z motion depends on the operator, the spring-damper response, and the equilibrium adjustment. A virtual plane or bounded corridor could make future scans more systematic.

## 2.4 Franka Research 3 as the Experimental Platform

The Franka Research 3 is a seven-degree-of-freedom manipulator designed for torque-controlled and contact-rich research [7], [8]. Its low-level interface provides high-rate access to torque control, robot state, and model information, which supports Cartesian impedance control and physical human-robot interaction. Kinematic redundancy also provides flexibility when selecting a joint configuration for the required end-effector pose.

The FR3 can estimate external joint torques and map them to an end-effector wrench. This estimate is useful for general interaction but is not equivalent to a probe-tip force measurement. Manual guidance, payload-model error, cables, tool inertia, and surface contact can all affect the same signal. This distinction governs the interpretation of the experimental results.

## 2.5 Contact-Force Sensing and Probe Orientation

Ultrasound contact control requires both a suitable measurement and a stable response. The target force must be sufficient for acoustic coupling while remaining below levels associated with discomfort, unsafe loading, or excessive deformation. The appropriate value depends on the probe, surface, coupling medium, and imaging task. Calibration, bias removal, bounded commands, and independent safety limits are therefore necessary.

A dedicated force sensor near the transducer can measure probe-surface interaction more directly [9], [20], [21]. Its mechanical load path remains important: a wrist sensor may detect both operator guidance and surface contact, whereas an axial load cell in the holder can better isolate the normal contact component. The planned next-stage hardware uses a Honeywell FSS force sensor with a PhidgetBridge interface [10], [11]. Filtering may reduce random noise, but it introduces delay and cannot separate forces that are already combined in the measurement.

A downward-facing probe is a practical initial orientation for scanning a horizontal phantom or body surface. It aligns the robot Z direction approximately with the surface normal and separates the main lateral scan direction from the axial contact direction without preventing compliant manual motion. The selected pose must still be checked for joint limits and poorly conditioned configurations throughout the intended workspace.

## 2.6 Motion Planning and Real-Time Interaction Control

MoveIt 2 is a ROS 2 framework for motion planning, manipulation, kinematics, collision checking, and trajectory execution [12]. These functions are appropriate when a desired pose or path can be specified before execution, and they remain relevant to future automatic coverage and collision-aware positioning. The present task instead requires commands to respond continuously to operator input and surface interaction. Because the operator remains inside the control loop, the complete scan path is not known in advance.

The Franka Control Interface provides a lower-level route for this interaction. A libfranka callback receives robot state and the elapsed control period, then returns a torque or motion command at a nominal 1 kHz rate [13]. Direct callback access does not make a controller inherently correct or safe, but it exposes the state, model, and torque-command structure needed for Cartesian impedance control. For this project, that transparency made the standalone example a more practical foundation than adding further planning and integration layers before the contact problem was understood.

The two architectures could be used together. ROS 2 and MoveIt could handle high-level positioning, scan planning, and system integration, while a real-time compliant controller manages local contact. This report focuses on the real-time controller because a planned ultrasound path is only useful if the probe can maintain reliable contact while following it.

## 2.7 Related Work and Position of This Project

Previous work includes purpose-built force-feedback robots, teleoperated systems, commercial collaborative platforms, and calibrated robotic 3D ultrasound acquisition [19]-[23]. These studies treat compliant motion, direct contact sensing, pose calibration, and system evaluation as related but distinct problems. One relevant Johns Hopkins system used separate sensing paths for operator guidance and probe-contact force [9]. That design influenced the sensor layout discussed here, although this project did not reproduce its platform or controller.

This project has a narrower implementation goal. It develops a human-guided controller on the available Franka platform and tests whether the robot's torque-derived wrench can support guidance and contact regulation at the same time. The controller could be operated as intended,

but the force estimate changed during manual interaction. This is the main reason for adding an external sensor and transducer holder in the next stage.

## 2.8 Chapter Summary

Autonomous robotic 3D ultrasound requires more than repeatable probe motion. The robot must associate each ultrasound frame with an accurate probe pose while maintaining suitable orientation and continuous contact with the patient. It must also regulate the probe-skin contact force to preserve acoustic coupling without applying excessive load or causing unnecessary tissue deformation. These requirements connect pose tracking and spatial calibration with real-time interaction control.

The controller addresses only an intermediate part of the full problem. The operator chooses the path, the robot can be repositioned by hand, and the Z equilibrium changes in response to the estimated contact force. The experiments indicate that direct force sensing is needed before contact regulation can be validated during motion. Probe integration and synchronized ultrasound acquisition should follow, with autonomous surface following and image-based adaptation considered later.

### 3. Project Development and Scope

#### 3.1 Initial Project Direction

The initial concept treated robotic 3D ultrasound as a complete acquisition pipeline. The FR3 would carry a tracked 2D transducer, execute a repeatable scan, associate each ultrasound frame with a robot pose, and support subsequent volume reconstruction. Early development therefore explored ROS 2, simulation, MoveIt, inverse kinematics, waypoint control, and Cartesian trajectory execution.

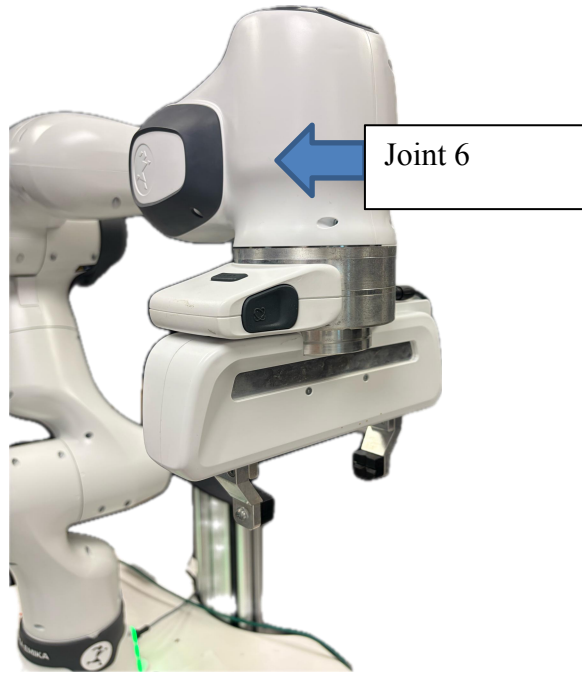
The early work showed how to generate planned robot motion, but it also exposed a more immediate problem. A predefined trajectory does not by itself let an operator reposition the probe naturally or maintain contact with an uncertain, compliant surface. Development therefore shifted to real-time physical interaction.

#### 3.2 Transition to Real-Time Interaction Control

ROS 2 compliant-control options were investigated, but their dependencies, configuration, hardware interfaces, and real-time integration introduced substantial overhead within the available project period. A more direct architecture was needed so that compliance, contact behaviour, and safety limits could be inspected and modified transparently.

The final implementation used a standalone controller based on the official libfranka Cartesian impedance example. With fewer software layers between the control law and the FR3, development could focus on impedance tuning, manual guidance, Z-equilibrium adjustment, orientation compliance, stopping behaviour, and experimental testing.

The controller evolved iteratively from v1 to v5.1. Early controller versions used a fixed lateral scan and comparatively strong pose constraints and gave way to compliant operator-guided motion, lower rotational resistance, reduced joint-6 resistance, controlled stopping, and reproducible data logging. XY remained the main scan plane, while Z remained manually adjustable through the impedance behaviour. This progression established the final experimental platform and exposed the limitation of using a torque-derived wrench during simultaneous human interaction and surface contact.



*Figure 3.1. Joints of the Franka Research 3, with Joint 6 highlighted in the wrist region near the end effector.*

### 3.3 Final System Concept and Objectives

The proposed complete system comprises the FR3 manipulator, an ultrasound transducer and dedicated holder, an axial force sensor near the end effector, the real-time robot controller, an ultrasound imaging system, and a data-acquisition computer. During scanning, the operator would guide the probe mainly laterally while retaining the ability to adjust its axial position through the compliant robot behaviour. The controller would adapt the desired axial equilibrium using the direct force measurement rather than restricting manual Z motion. Time-aligned robot poses and ultrasound frames would then support 3D reconstruction.

The current prototype implements the robot-control and logging portions of this architecture. It reads the FR3 state and model terms, constructs a bounded Cartesian task wrench, maps that wrench to joint torque through the Jacobian transpose, and records selected variables at approximately 100 Hz. Figure 3.2 separates the implemented controller from the force sensing, ultrasound, calibration, and reconstruction stages that remain future work.

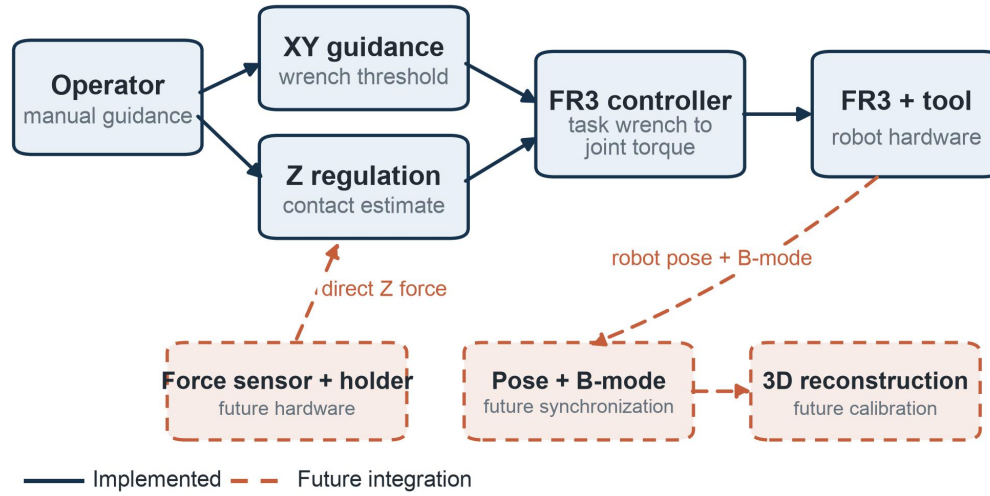


Figure 3.2. Proposed human-guided robotic 3D ultrasound system and information flow.

Table 3.1 summarizes the project objectives and distinguishes implemented controller functions and experimental observations from hardware integration and imaging stages that were not completed.

Table 3.1. Project objectives, status, and evaluation evidence.

| No. | Objective                                                                              | Status and evidence                                                                                                   |
|-----|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 1   | Establish Cartesian impedance control on the Franka robot.                             | Implemented from the libfranka example and exercised in the reported interaction tests.                               |
| 2   | Enable compliant human-guided translational motion during scanning.                    | Substantial manually guided displacement was demonstrated; Z remained manually adjustable through impedance.          |
| 3   | Apply contact-related Z-equilibrium adaptation around a selected target.               | Implemented without rigidly constraining manual Z motion; evaluated using the internal force estimate.                |
| 4   | Maintain manually adjustable end-effector orientation for future probe mounting.       | Implemented qualitatively through low rotational stiffness and joint-6 task-torque scaling.                           |
| 5   | Provide software safety limits and controlled termination.                             | Software limits and controlled stopping were implemented; clinical safety was not validated.                          |
| 6   | Record data required for quantitative evaluation.                                      | Timestamped CSV logging was implemented at approximately 100 Hz.                                                      |
| 7   | Identify the limitation of torque-based force estimation during cooperative motion.    | No-contact guidance trials showed that operator interaction altered the estimated Z contact force.                    |
| 8   | Prepare the controller for direct force sensing and robotic 3D ultrasound acquisition. | External sensing, holder integration, calibration, ultrasound synchronization, and reconstruction remain future work. |

### **3.4 Scope and Limitations**

The completed scope includes v1-v5.1 controller development, compliant-interaction tuning, software limits, controlled stopping, data logging, and laboratory evaluation of manual motion, Z-equilibrium adaptation, and torque-derived contact estimation. It does not establish true probe-force accuracy or clinical safety. The external sensor and transducer holder were not integrated, and synchronized ultrasound acquisition and 3D reconstruction were not performed. Because the parameters were tuned only for the available setup, they cannot be assumed to transfer to a mounted probe, another surface, or a human subject. Direct force-sensor integration and calibration are required before ultrasound acquisition begins.



## 4. Theoretical Framework

This chapter presents the theory underlying the final real-time interaction controller. Motion planning, inverse kinematics, and waypoint execution informed early development but are not part of the inner loop evaluated here. The v5.1 program instead maps measured FR3 state and model quantities directly to Cartesian task forces and joint torques. The discussion follows the implemented design: robot kinematics and dynamics, Cartesian impedance, primarily XY scan guidance, shared manual and automatic Z behaviour, orientation compliance, and the limitations of the torque-derived external wrench.

### 4.1 Coordinate Frames and Differential Kinematics

The controller operates in both joint space and Cartesian task space. The joint-position vector  $q$  contains the seven FR3 joint angles, while the end-effector pose contains its position and orientation relative to the robot base. The geometric Jacobian  $J(q)$  maps joint velocity to the six-dimensional Cartesian velocity of the end effector:

$$\dot{x} = J(q)\dot{q} \quad (4.1)$$

The first three components of the Cartesian velocity in (4.1) describe translation, and the final three describe angular motion. In the implemented controller, X and Y form the primary scan plane and Z is treated as the nominal surface-normal direction. These labels indicate the main function of each direction rather than restricting operator motion. Cartesian impedance allows manual displacement in Z as well as in XY. The convention is most intuitive when the end effector initially points downward toward a horizontal test surface.

The robot base, end-effector, force-estimate, and future probe-image frames must remain distinct. The v5.1 controller uses the end-effector pose  $OT_{EE}$  and the estimated external wrench  $\widehat{OF}_{ext,K}$  supplied in the Franka robot state. A complete ultrasound system will also require a calibrated transformation from the end effector, through the force sensor and holder, to the ultrasound image plane.

## 4.2 Robot Dynamics and Cartesian Torque Control

The dynamics of a serial manipulator can be represented using the inertia matrix, Coriolis and centrifugal effects, gravity, commanded joint torque, and external joint torque. A general form is:

$$M(q)\ddot{q} + c(q, \dot{q}) + g(q) = \tau + \tau_{ext} \quad (4.2)$$

The controller does not explicitly solve this full equation at every cycle. It constructs a desired Cartesian task wrench and maps that wrench to joint torque using the transpose of the Jacobian, consistent with an operational-space formulation [15]. The Coriolis term returned by the Franka model is then added to the command:

$$\tau_d = J(q)^T W_{task} + c(q, \dot{q}) \quad (4.3)$$

Gravity compensation is handled internally by the FR3 low-level controller and is therefore not added explicitly to the commanded torque in (4.3). This formulation allows translational forces and rotational moments to be designed in Cartesian coordinates while the robot receives a seven-element joint-torque command. The seventh degree of freedom provides redundancy, although the selected configuration must still avoid joint limits and poorly conditioned Jacobians.

## 4.3 Cartesian Impedance and Damping

Cartesian impedance control creates a virtual mechanical relationship between the actual and desired end-effector states. For one translational direction, a spring-damper force can be written as:

$$f = -K(x - x_d) - D\dot{x} \quad (4.4)$$

The stiffness  $K$  determines how strongly the controller resists displacement from equilibrium, while  $D$  opposes motion and suppresses oscillation [14], [16]. The implementation uses the example-code choice  $D = 2\sqrt{K}$ . This expression assumes normalized unit effective inertia and is not a dimensionally complete critical-damping model of the robot-environment system. Actual damping depends on configuration-dependent Cartesian inertia, environment stiffness, sampling, saturation, and model behaviour. In v5.1, the spring-damper form is retained in  $Z$  and rotation.  $XY$  stiffness is removed so that the robot does not return to a fixed horizontal

position during manual guidance. The finite Z stiffness still permits axial displacement while providing a compliant restoring response around a movable equilibrium.

Table 4.1. Functional control roles by task direction.

| Component       | Primary input                                  | Control behaviour                                                            | Purpose                                                                       |
|-----------------|------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| X-Y translation | Bias-corrected XY wrench-estimate direction    | Fixed desired speed with Cartesian velocity damping                          | Permit lateral scan guidance and generate a bounded velocity-related command. |
| Z translation   | Manual axial input and estimated contact force | Compliant spring-damper response with force-dependent equilibrium adaptation | Preserve manual Z adjustment while superimposing contact-related adaptation.  |
| Rotation        | Orientation error and angular velocity         | Low-stiffness rotational impedance                                           | Keep the tool stable but manually adjustable.                                 |
| Joint 6         | Cartesian task torque                          | Scaled task-torque contribution                                              | Reduce resistance to manual end-effector rotation.                            |

#### 4.4 Human-Guided XY Velocity Control

The robot-state wrench is first corrected using the force measured at the initial stationary pose. For each Cartesian component  $i$ , the bias-corrected force is:

$$F_{i,corr} = F_{i,raw} - F_{i,bias} \quad (4.5)$$

The magnitude of the bias-corrected XY wrench estimate is calculated from its X and Y components. Below a threshold, the desired horizontal velocity is zero. Above the threshold, only the estimated wrench direction is used and the desired speed has a fixed command magnitude:

$$F_{xy} = \sqrt{F_{x,corr}^2 + F_{y,corr}^2} \quad (4.6)$$

$$v_{xy,d} = \begin{cases} 0 & \text{if } F_{xy} \leq F_{xy,th} \\ v_{cmd} \frac{[F_{x,corr} \ F_{y,corr}]^T}{F_{xy}} & \text{if } F_{xy} > F_{xy,th} \end{cases} \quad (4.7)$$

This direction-only mapping prevents a large applied force from directly commanding an arbitrarily large speed. The actual horizontal velocity is obtained from the Jacobian relation in

(4.1). Cartesian velocity damping generates the XY task force required to approach the desired velocity:

$$f_{xy} = -D_{xy}(v_{xy} - v_{xy,d}) \quad (4.8)$$

The XY task-force norm is saturated before entering the task wrench. The controller also integrates the desired XY velocity to maintain a reference position for logging, but this reference is not used as a horizontal spring equilibrium. Operator interaction selects the direction, and the controller contributes a bounded velocity-related force.

#### 4.5 Z-Direction Contact Regulation

Because the tool points nominally downward, compressive contact was assumed to correspond to a negative bias-corrected base-frame Z force. The controller converts this signed component to the non-negative estimate  $\hat{F}_{contact} = \max(-F_{z,corr}, 0)$ . This rectification is a control convention rather than a direct force measurement. Positive  $F_{z,corr}$  values become zero, so their magnitude is discarded.

$$\hat{F}_{contact} = \max(-F_{z,corr}, 0) \quad (4.9)$$

The force error is the difference between the selected target and the estimated contact force. The v5.1 controller does not scale equilibrium speed continuously with error magnitude. Once the estimate exceeds the contact threshold and lies outside the deadband, the desired Z equilibrium moves at a fixed speed in the direction that reduces the error:

$$e_F = F_{target} - \hat{F}_{contact} \quad (4.10)$$

$$\dot{z}_d = \begin{cases} -v_z \text{sgn}(e_F) & \text{if } \hat{F}_{contact} > F_{contact,th} \wedge |e_F| > \Delta F_{deadband} \\ 0 & \text{otherwise} \end{cases} \quad (4.11)$$

Under the coordinate convention used in the code, a positive force error moves the desired equilibrium downward, whereas a negative error moves it upward. The desired height is bounded around the initial pose. This value is an impedance equilibrium rather than a rigid position command, so the operator can still move the end effector above or below it by applying axial

force. The resulting Z task force combines the spring-damper response to manual displacement with the contact-related movement of the equilibrium and is saturated separately:

$$f_z = \text{clamp} \left( -K_z (z - z_d) - D_z \dot{z}, -f_{z,max}, f_{z,max} \right) \quad (4.12)$$

The contact threshold prevents equilibrium adaptation before contact is detected, and the deadband reduces repeated switching near the target. A separate maximum-force condition terminates control when the estimated contact force exceeds the software limit.

## 4.6 Orientation Control and Joint 6 Compliance

Unit quaternions represent end-effector orientation without the parameter singularities of Euler angles and support feedback control using a compact orientation-error representation [18]. Let  $Q$  and  $Q_d$  denote the measured and desired orientations, and let  $R$  denote the measured end-effector rotation matrix. The desired quaternion follows the manually guided orientation gradually through spherical linear interpolation. The code uses the following quaternion-error convention and rotational impedance moment:

$$\begin{aligned} e_R &= -R \text{vec} (Q^{-1} Q_d) \\ m_R &= -K_R e_R - D_R \omega \end{aligned} \quad (4.13)$$

Low rotational stiffness allows manual tool adjustment, while damping limits rapid angular motion. The rotational moment is norm-limited before it is added to the task wrench. To reduce resistance during manual rotation, the Cartesian task-torque contribution at joint 6 is scaled while Coriolis compensation is retained:

$$\tau_{d,6} = c_6 + s_6 \tau_{task,6} \quad 0 \leq s_6 \leq 1 \quad (4.14)$$

A scale of one preserves the full task torque and a smaller scale reduces resistance at that joint. In v5.1, the separate joint-6 locking stiffness and locking torque limit are set to zero, so the principal joint-6 modification is the Cartesian task-torque scaling.

## 4.7 Torque-Based Force Estimation and Its Limitation

In principle, an end-effector wrench and external joint torques are related through the Jacobian transpose:

$$\tau_{ext} \approx J(q)^T W_{ext} \quad (4.15)$$

The v5.1 controller does not compute a wrench inverse. Instead, it reads the model-based external-wrench estimate supplied in the Franka robot state. This API quantity is associated with stiffness frame  $K$  and follows the Franka frame convention; it is not a calibrated probe-tip force. The estimate combines external interaction with effects caused by model mismatch. Subtracting a single initial sample removes the offset at that instant but cannot identify the source of a later change or correct configuration-dependent drift.

During manual operation, the estimated wrench can be affected by XY guidance, axial hand forces, tool and cable loads, and surface contact. Its Z component may therefore change even when the actual probe contact does not. A low-pass or Kalman filter could smooth random noise, but it could not separate inputs that are already mixed in the same estimate. The main problem is the source of the measurement, not simply signal noise.

#### 4.8 Hybrid Controller and Safety Structure

The complete task wrench combines XY velocity damping, Z impedance force, and rotational impedance moment:

$$W_{task} = [f_x f_y f_z m_x m_y m_z]^T \quad (4.16)$$

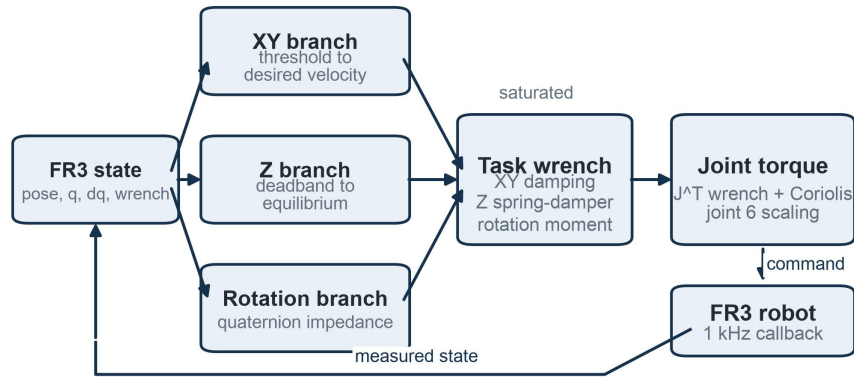


Figure 4.1. Theoretical structure of the v5.1 hybrid controller.

The controller limits XY speed, Z-equilibrium motion, Cartesian force, rotational moment, and the contact estimate. Franka collision behaviour remains active, and Enter requests controlled termination. Model-based collision detection and controlled reaction are established components of safe interaction with lightweight manipulators [24], while broader pHRI guidance also emphasizes system-level risk reduction and intuitive interaction [25]. The present laboratory safeguards do not constitute clinical safety validation.

## 5. Methodology and System Implementation

### 5.1 Development Path and Control-Architecture Selection

Chapter 3 described the shift from planned scanning to real-time interaction. The implemented controller required direct access to FR3 state, model terms, and torque commands so that compliance, contact-related adaptation, limits, and logging could be evaluated transparently.

The official libfranka Cartesian impedance example was selected as a traceable implementation base. ROS 2 and MoveIt remain relevant to future high-level positioning and system integration, while v5.1 focuses on local real-time interaction and buffered data logging.

### 5.2 Experimental Platform and Current System Boundary

The experimental platform was a Franka Research 3, a seven-degree-of-freedom collaborative robot with joint-torque sensing and a low-level torque-control interface. The workstation ran Ubuntu 24.04.3 LTS with the 6.8.1-1035-realtime kernel. ROS 2 Jazzy was available in the workspace, but the v5.1 controller was a standalone C++ program built with libfranka 0.13.6-1 and Eigen. It communicated directly with the robot using the hostname supplied as a command-line argument and was not implemented as a ROS 2 controller plugin. The compiler version, Eigen version, FR3 system-software version, and controller-binary build date were not recorded during the experiments.

The completed experiments evaluated robot interaction and control. The ultrasound transducer holder and external force sensor were not integrated during the principal v5.1 trials, so contact force was represented by the FR3 model-based external-wrench estimate rather than a direct probe-tip measurement. The future system will add the transducer, holder, force sensor, ultrasound system, and image-pose synchronization. These components belong to the system design but not to the completed experimental hardware.





*Figure 5.1. FR3 experimental platform and current laboratory arrangement.*

### 5.3 Software Architecture and Initialization

The final program retained the structure of the libfranka Cartesian impedance example while replacing its fixed-pose behaviour with the human-assisted hybrid interaction controller. At startup, the program connects to the robot, applies the configured default behaviour, loads the FR3 dynamics model, and calls `robot.readOnce()` to obtain an initial state. The X, Y, and Z components of the external-wrench estimate from this single state become static bias values. The same state initializes the desired position and orientation.

The program then configures collision behaviour and waits for the operator to press Enter. The first input starts torque control and data recording. A separate input thread waits for a second Enter input and sets an atomic stop flag. The real-time callback detects the flag, returns the final computed torque command through `MotionFinished`, and exits. If the rectified estimate exceeds 20 N, the supplied code instead requests `MotionFinished` with a seven-element zero-torque command. The application code does not implement a dedicated torque ramp for this overload path, and its transient response was not characterized experimentally. Buffered samples are written to a timestamped CSV file after control ends. Neither stop path was validated as fail-safe or medical-device behaviour.

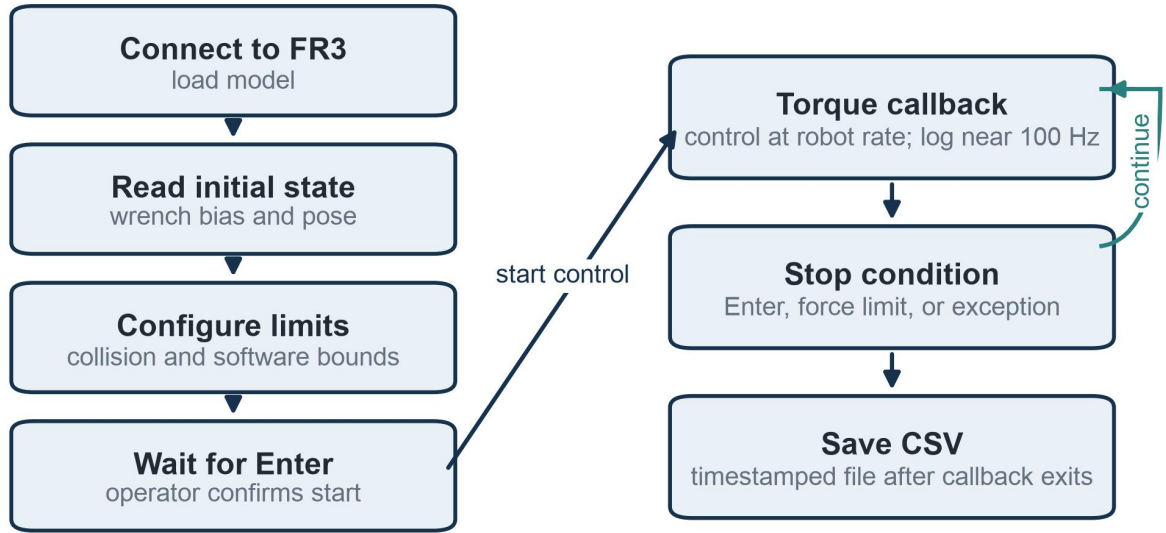


Figure 5.2. Software execution sequence for the v5.1 controller.

## 5.4 Iterative Controller Development

The controller was developed iteratively, with each version addressing an interaction problem observed in the previous one. Table 5.1 summarizes this development path. The version labels indicate project order and should not be interpreted from file-modification timestamps.

Table 5.1. Development of the controller from v1 to v5.1.

| Version | Principal implementation                                                                                                          | Reason for change                                                               | Outcome                                                                                       |
|---------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| v1      | Cartesian impedance, force-bias calibration, Z equilibrium adjustment, slow X scan, and joint 6 locking.                          | Establish basic contact regulation and repeatable lateral motion.               | Provided the initial force-control prototype but remained restrictive for manual interaction. |
| v2      | Reduced stiffness, removed joint 6 locking, added XY hand guidance, and allowed orientation following.                            | Make the robot easier to reposition and rotate by hand.                         | Improved compliance but retained position-oriented guidance behaviour.                        |
| v3      | Changed to direction-based XY velocity guidance, retained Z contact regulation, and added velocity damping and task-force limits. | Combine scan-plane guidance with shared manual and contact-related Z behaviour. | Established the main compliant guidance and Z-equilibrium-adaptation architecture.            |
| v4      | Reduced XY activation threshold and limits, adjusted XY guidance response, and increased Z response.                              | Increase responsiveness during physical guidance and contact tests.             | Made manual commands easier to trigger but required further speed tuning.                     |
| v5      | Increased XY command speed,                                                                                                       | Improve practical scan motion                                                   | Produced the final                                                                            |

| Version | Principal implementation                                                                      | Reason for change                                                            | Outcome                                                                         |
|---------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
|         | reduced rotational resistance, and scaled Cartesian task torque at joint 6.                   | and manual end-effector rotation.                                            | interaction behaviour used for evaluation.                                      |
| v5.1    | Added buffered 100 Hz logging, timestamped filenames, and Enter-based controlled termination. | Collect reproducible report data without file I/O in the real-time callback. | Enabled quantitative post-processing while preserving the v5 control structure. |

## 5.5 Final v5.1 Control-Cycle Implementation

The FR3 torque callback receives the current robot state and elapsed control period. The implementation performs the operations in Table 5.2 during each cycle. This ordering ensures that measured state and model quantities are available before the Cartesian task wrench and joint torques are constructed.

*Table 5.2. Operations performed in the v5.1 torque-control callback.*

| Step | Operation                                                                                                                                                        | Implementation output                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 1    | Read raw external wrench and subtract the startup bias.                                                                                                          | Corrected $F_x$ , $F_y$ , and $F_z$ .                |
| 2    | Convert negative corrected $F_z$ into a non-negative contact-force estimate and check the force limit.                                                           | Contact estimate or force-limit termination request. |
| 3    | Read Coriolis terms, zero Jacobian, joint state, and end-effector pose from the FR3 model and state.                                                             | Model and kinematic quantities.                      |
| 4    | Map joint velocity through the Jacobian and determine the XY command from the bias-corrected wrench estimate.                                                    | Actual Cartesian velocity and desired XY velocity.   |
| 5    | Apply Cartesian velocity damping and XY task-force saturation.                                                                                                   | Bounded XY task force.                               |
| 6    | Update desired orientation toward the hand-guided orientation using quaternion interpolation.                                                                    | Compliant desired orientation.                       |
| 7    | Use the contact threshold, deadband, and force-error sign to update desired Z position.                                                                          | Bounded Z equilibrium point.                         |
| 8    | Calculate and limit the Z spring-damper force.                                                                                                                   | Bounded Z task force.                                |
| 9    | At the logging interval, copy selected variables into the preallocated memory buffer.                                                                            | One LogSample record.                                |
| 10   | Calculate and limit rotational impedance, assemble the Cartesian wrench, apply the Jacobian transpose, add Coriolis compensation, and scale joint 6 task torque. | Seven commanded joint torques.                       |

## 5.6 Parameter Selection and Tuning

Parameters were selected empirically through repeated low-speed interaction tests. Tuning aimed to reduce resistance during XY guidance and manual rotation while retaining damping, bounded task forces, and a detectable Z-contact response. Table 5.3 lists the values used in the reported v5.1 experiments. They describe the tested software configuration, not clinically validated settings.

*Table 5.3. Principal v5.1 controller parameters.*

| Parameter                      | Code value  | Unit    | Function                                                          |
|--------------------------------|-------------|---------|-------------------------------------------------------------------|
| XY force threshold             | 1.0         | N       | Activates human-guided XY motion.                                 |
| XY command speed               | 0.005       | m/s     | Sets the fixed desired lateral speed.                             |
| XY velocity damping            | 30.0        | N s/m   | Regulates actual XY velocity toward the desired velocity.         |
| Z stiffness                    | 18.0        | N/m     | Creates the Z virtual spring.                                     |
| Z damping                      | 8.49        | N·s/m   | Provides nominal damping for Z motion.                            |
| Target contact force           | 3.0         | N       | Defines the desired torque-derived contact estimate.              |
| Contact threshold              | 1.0         | N       | Prevents Z adaptation before contact detection.                   |
| Force deadband                 | 1.0         | N       | Reduces switching near the target force.                          |
| Z equilibrium speed            | 0.02        | m/s     | Configured fixed rate for Z-equilibrium adaptation.               |
| Maximum Z offset               | 0.3         | m       | Configured bound around the initial Z equilibrium position.       |
| Estimated-force stop limit     | 20.0        | N       | Terminates control above the software force limit.                |
| Rotational stiffness           | 0.5         | N·m/rad | Maintains low-stiffness orientation behaviour.                    |
| Angular follow speed           | 1.0         | rad/s   | Updates desired orientation toward measured orientation.          |
| Maximum XY / Z task force      | 10.0 / 10.0 | N       | Bounds translational task forces.                                 |
| Maximum rotational task moment | 3.0         | N·m     | Bounds the rotational command.                                    |
| Joint 6 task-torque scale      | 0.3         | -       | Retains 30% of the Cartesian task-torque contribution at joint 6. |

| Parameter                      | Code value                   | Unit                                            | Function                                                                |
|--------------------------------|------------------------------|-------------------------------------------------|-------------------------------------------------------------------------|
| Rotational damping             | 1.41                         | $\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$ | Provides nominal damping for orientation motion.                        |
| Joint collision thresholds     | 80 for all joints            | $\text{N}\cdot\text{m}$                         | Configures the development collision reflex thresholds.                 |
| Cartesian collision thresholds | 35 translation / 20 rotation | $\text{N} / \text{N}\cdot\text{m}$              | Configures translational-force and rotational-moment reflex thresholds. |

The v5.1 experiments used a Z-equilibrium speed of 0.02 m/s (20 mm/s) and a maximum Z offset of 0.3 m (300 mm). The latter was a broad development bound, not an application-specific safe travel range. Z damping was 8.49  $\text{N}\cdot\text{s}/\text{m}$  and rotational damping was 1.41  $\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$ . Collision behaviour used joint-torque thresholds of 80  $\text{N}\cdot\text{m}$  for all joints and Cartesian thresholds of 35 N in translation and 20  $\text{N}\cdot\text{m}$  in rotation. These thresholds permitted laboratory interaction but were not validated as human-contact or medical safety limits.

## 5.7 Data Logging and File Management

The libfranka callback runs at the robot control rate, while report variables are sampled every 0.01 s, or approximately 100 Hz. To avoid file operations inside the real-time callback, the program reserves memory for 120,000 LogSample structures and appends a sample when the logging interval has elapsed. This capacity corresponds to approximately 20 minutes at 100 Hz. After controlled termination, the samples are written to CSV with six decimal places.

Each file is stored in /home/vasst/impedance control log and named robot\_control\_log\_YYYYMMDD\_HHMMSS.csv. Timestamped filenames prevent a new test from overwriting an earlier run. The program reports an error if the destination file cannot be opened and prints the number of saved samples and final path after a successful save.

Table 5.4. Logged variables and their intended use in the report.

| Variable group       | CSV channels                             | Intended analysis                                     |
|----------------------|------------------------------------------|-------------------------------------------------------|
| Time                 | time                                     | Time axis, duration, and effective sampling interval. |
| Raw force estimate   | raw_fx, raw_fy, raw_fz                   | Document the original FR3 wrench estimate.            |
| Bias-corrected force | corrected_fx, corrected_fy, corrected_fz | Assess startup calibration and interaction signals.   |

| Variable group    | CSV channels                               | Intended analysis                                                                                    |
|-------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------|
| Contact variables | contact_force, target_force, force_xy_norm | Compare the rectified internal estimate with its nominal target and inspect XY interaction coupling. |
| XY velocity       | v_des_x, v_des_y, v_x, v_y                 | Identify guidance activation and summarize measured horizontal motion.                               |
| Z motion          | z_dot, z, z_d, z_offset, f_z               | Evaluate equilibrium adaptation, vertical motion, and commanded task force.                          |
| Pose reference    | x, y, x_d, y_d                             | Plot the horizontal trajectory and integrated guidance reference.                                    |

## 5.8 Preliminary Experimental Procedure

Controller data were collected on 30 July 2026 under four principal conditions, Pose 1 through Pose 4, plus an additional forward/backward guidance condition. Every run used v5.1 and a downward-facing end-effector configuration. Pose 2 and Pose 4 used a soft, skin-like test surface. Before torque control began, one robot state was recorded for force-bias subtraction; the implementation did not average a stationary calibration interval. The operator pressed Enter once to begin torque control and logging and again to request termination. Each timestamped CSV file was associated with its condition through the filename or laboratory notes.

*Table 5.5. Experimental conditions and completion status.*

| Condition                | Procedure                                                                                                                                | Trials and status                                                               |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Pose 1                   | Hold the FR3 stationary without intended surface contact to characterize the corrected baseline.                                         | Three trials completed.                                                         |
| Pose 2                   | Maintain nominally stationary contact with a soft, skin-like test surface and observe the torque-derived estimate around the 3 N target. | Seven logs recorded; four steady-state intervals selected for primary analysis. |
| Pose 3                   | Guide the FR3 laterally in XY without intended surface contact.                                                                          | Three trials completed.                                                         |
| Pose 4                   | Guide the FR3 laterally while maintaining contact with the same soft, skin-like test surface; no ultrasound images were acquired.        | Two preliminary trials completed.                                               |
| Directional guidance     | Move manually along the same approximate X corridor in two backward and two forward trials.                                              | Four trials completed; force-estimate coupling analysed separately.             |
| Direct-sensor comparison | Repeat selected conditions after the external sensor and holder are integrated and calibrated.                                           | Not completed; future work.                                                     |

The dataset comprised 19 trials: three Pose 1 baseline logs, seven Pose 2 stationary-contact logs, three Pose 3 general-guidance logs, two Pose 4 combined contact-and-guidance logs, and four additional directional-guidance logs. Four Pose 2 intervals were selected retrospectively for the primary stationary analysis. Selection was not algorithmic: an interval was retained when it followed initial contact establishment and did not contain a large Z transient, repeated loss of the rectified estimate below the 1 N contact threshold, or unintended XY-guidance activation. Trial 135614 was excluded because of an approximately 57 mm initial Z transient, trial 135933 contained substantial XY activation during the nominally stationary condition, and trial 150749 contained repeated contact loss. Exact selected windows are reported in Appendix A. Because the logs contained no manual event markers, this procedure is an exploratory post hoc analysis rather than a prospective acceptance test. Direct-sensor comparison remains future work until the sensor load path, calibration, sampling rate, and timing have been verified.

## 5.9 Safety Measures and Methodological Limitations

Laboratory trials used bounded XY speed and Z-equilibrium motion, saturated Cartesian task forces and moments, Franka collision behaviour, force-limit and Enter-based stop paths, and access to the robot user stop. The configured collision thresholds and 0.3 m Z-equilibrium range were development settings rather than application-specific safety limits. The overload path requested a zero-torque MotionFinished command without a dedicated application-level ramp, and its stopping transient was not characterized. Model-based detection and post-collision reaction have been studied for lightweight manipulators [24], but implementing comparable principles does not by itself establish safety for a medical task. Pose 4 was a limited soft-surface test without ultrasound acquisition or independent force measurement. Future work should use a fixed phantom after direct sensing, conservative travel limits, collision thresholds, and stopping behaviour have been validated. The present measures do not constitute medical-device or clinical safety validation.

The main methodological limitation is that the same FR3 torque-derived wrench was used for guidance detection and Z-contact regulation. Because this estimate does not isolate surface force, the experiments evaluate controller behaviour and sensing limitations rather than true contact-force accuracy. Other limitations include empirical tuning, the absence of synchronized

ultrasound frames and 3D reconstruction, no manual event markers, and a small number of trials. These constraints define the interpretation of Chapter 6.



## 6. Results and Discussion

### 6.1 Experimental Overview and Analysis Method

Nineteen v5.1 logs were collected on 30 July 2026: three stationary no-contact trials (Pose 1), seven nominally stationary soft-surface contact trials (Pose 2), three human-guided no-contact trials (Pose 3), two combined guidance-and-contact trials (Pose 4), and four additional directional-guidance trials. Table 6.1 summarizes the conditions. The analysis addressed three questions: whether an operator could move the FR3 compliantly, whether the torque-derived contact estimate remained stable during stationary operation, and whether that estimate remained interpretable during manual guidance. No ultrasound images were acquired.

*Table 6.1. Preliminary experimental dataset.*

| Condition            | Trials                 | Mean duration                          | Intended interaction                                | Interpretation                                                            |
|----------------------|------------------------|----------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------|
| Pose 1               | 3                      | $13.5 \pm 3.0$ s                       | Stationary, no surface contact                      | Baseline of the torque-derived signal.                                    |
| Pose 2               | 7 total;<br>4 selected | $19.8 \pm 8.5$ s<br>selected intervals | Stationary soft-surface contact                     | Steady torque-derived contact regulation; transients reported separately. |
| Pose 3               | 3                      | $18.2 \pm 2.1$ s                       | Human-guided XY, no intended contact                | Tests manual guidance and apparent contact-estimate coupling.             |
| Pose 4               | 2                      | $31.6 \pm 13.3$ s                      | Human-guided XY with soft-surface contact           | Combined guidance and soft-surface contact test.                          |
| Directional guidance | 4                      | $14.1 \pm 2.4$ s                       | Two backward and two forward manual-guidance trials | Interaction-dependent coupling in the torque-derived force estimate.      |

The median logging interval was 10-11 ms, corresponding to approximately 91-100 Hz. Four Pose 2 intervals were retained using the retrospective rule defined in Section 5.8; Appendix A lists the exact windows. Results were first calculated within each trial and then summarized across trial-level values, preventing longer recordings from being treated as additional repetitions. Given the exploratory protocol and small sample, the analysis uses descriptive statistics only. No additional time-domain filtering was applied.

## 6.2 Stationary Behaviour

During Pose 1, the no-contact torque-derived estimate averaged  $0.29 \pm 0.11$  N across the three trial means, with a mean within-trial standard deviation of 0.03 N. The estimate remained below the 1 N contact threshold in all three recordings. The small positive residual is consistent with single-sample startup bias subtraction, configuration-dependent model error, and rectification of the corrected Z component.

Across the four selected Pose 2 steady-state intervals, the torque-derived contact estimate averaged  $3.05 \pm 0.12$  N for a nominal 3 N target. The mean within-trial standard deviation was  $0.16 \pm 0.03$  N, and the trial-level RMSE relative to the nominal internal target was  $0.19 \pm 0.07$  N. Figure 6.1 presents the four intervals, and Table 6.2 summarizes the descriptive results.

During the selected steady intervals, the torque-derived estimate stayed near the target when the robot was relatively undisturbed. This shows that the controller regulated its own estimate; it does not show that the true surface force was 3 N. There was no independent force sensor, the intervals were chosen after the experiment, and some complete recordings contained startup transients or loss of contact. The stationary results should therefore be treated as a limited controller test, not a validation of force accuracy.

*Table 6.2. Descriptive contact-force results by experimental condition. Values are mean  $\pm$  SD across trial-level summaries; the within-trial SD column summarizes the trial-specific standard deviations.*

| Condition                                       | Mean estimate (N) | Within-trial SD (N) | RMSE relative to nominal internal target (N) | Samples at or below 1 N (%) |
|-------------------------------------------------|-------------------|---------------------|----------------------------------------------|-----------------------------|
| Pose 1: no contact                              | $0.29 \pm 0.11$   | $0.03 \pm 0.01$     | Not meaningful                               | $100.00 \pm 0.00$           |
| Pose 2: retrospectively selected steady contact | $3.05 \pm 0.12$   | $0.16 \pm 0.03$     | $0.19 \pm 0.07$                              | $0.00 \pm 0.00$             |
| Pose 3: XY, no contact                          | $1.42 \pm 0.31$   | $1.77 \pm 0.16$     | Not meaningful                               | $59.50 \pm 7.42$            |
| Pose 4: soft-surface contact + XY               | $2.54 \pm 0.08$   | $0.90 \pm 0.20$     | $1.02 \pm 0.14$                              | $5.38 \pm 0.14$             |

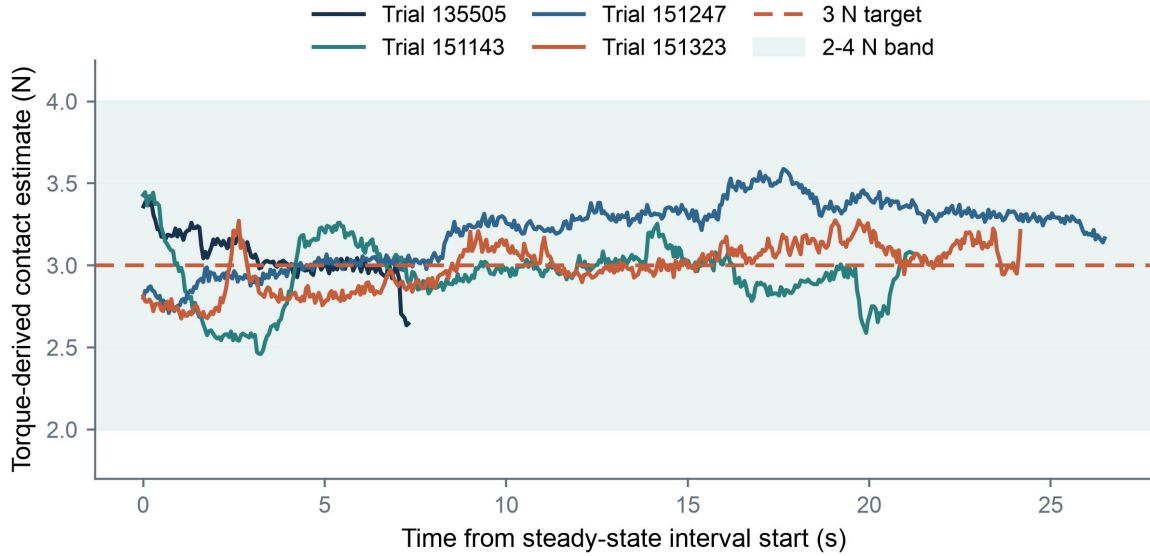


Figure 6.1. Torque-derived contact-force estimate during the four selected Pose 2 steady-state intervals. Time is expressed relative to the beginning of each analysed interval; the dashed line is the 3 N target and the shaded band is 2-4 N.

### 6.3 Human-Guided Motion and Force-Estimate Coupling

Across the three Pose 3 trials, the X range was  $440 \pm 18$  mm and the Y range was  $23 \pm 5$  mm. As shown in Figure 6.2, the operator produced mainly horizontal sweeps without a pre-planned path. The trajectories confirm that the robot could be moved through a substantial Cartesian distance by hand.

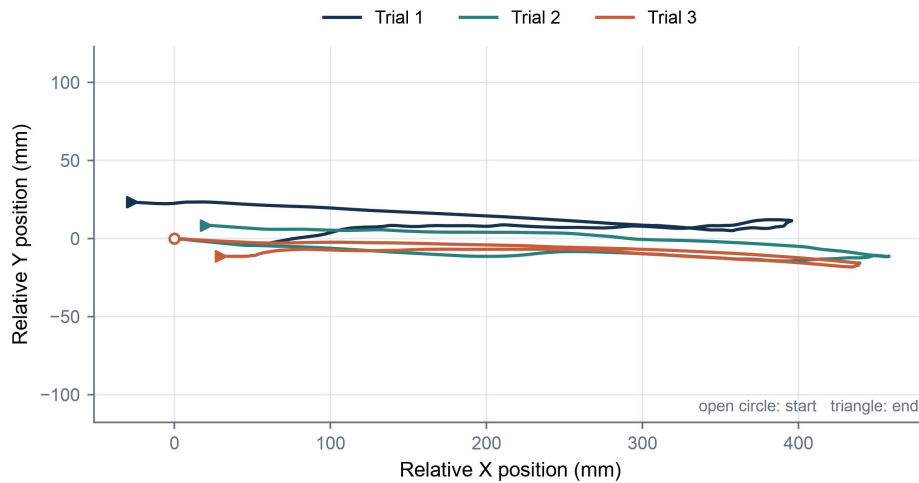


Figure 6.2. Relative XY trajectories for the three Pose 3 human-guidance trials. Each path is expressed relative to its own initial position and displayed with equal spatial scaling.

XY guidance was active for most of each Pose 3 recording, and the measured XY speed during those periods was approximately  $0.050 \pm 0.003$  m/s. Most of this speed came from the

operator physically moving the compliant robot, so it should not be read as velocity-tracking performance. It could also exceed the 0.005 m/s guidance reference because that reference was not a hard limit on manual displacement.

Pose 3 showed the main sensing problem. Although the robot was not intended to touch a surface, the torque-derived contact estimate averaged  $1.42 \pm 0.31$  N across trial means, reached 6.64 N, and exceeded the 1 N contact threshold in 40.5% of samples. Figure 6.3 shows one trial. In this condition, the controller sometimes interpreted the operator's input as Z contact.

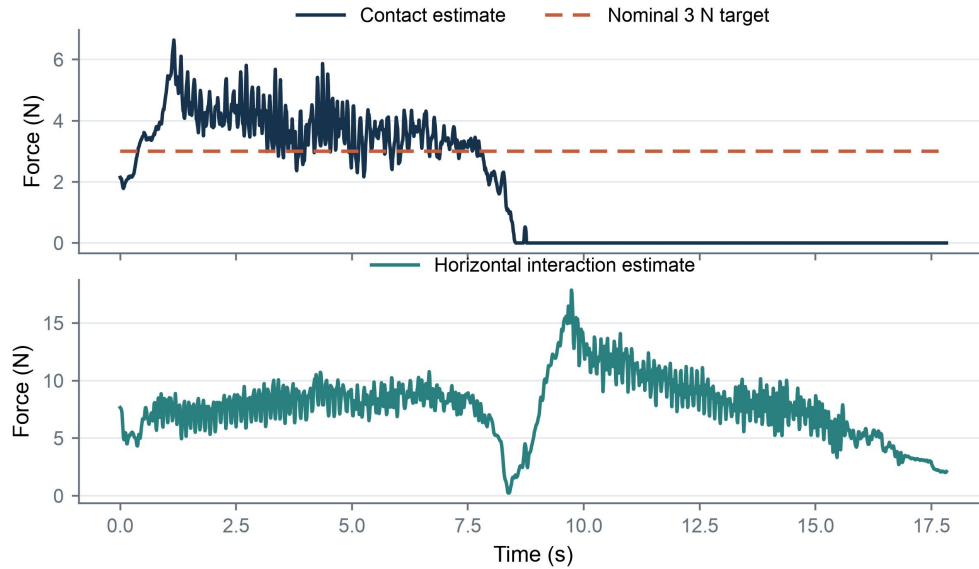


Figure 6.3. Torque-derived contact estimate and horizontal interaction estimate during representative no-contact manual guidance. The 3 N line is the controller's nominal target, not a measured ground truth.

This matches the behaviour seen during manual guidance, when the displayed Z estimate increased or decreased as the operator pushed the robot. The response also changed with robot configuration and with how the external load passed through the joints. The torque-derived wrench could not uniquely identify surface contact.

The four directional trials provide a second illustration of this coupling. Two trials in one travel direction produced mean rectified estimates of 0.01 N and 0.04 N, while two trials in the opposite direction produced means of 3.62 N and 2.95 N and a maximum of 7.69 N. Figure 6.4 shows this asymmetry. Because hand-force magnitude, Z displacement, initial bias, robot configuration, and dynamic state were not controlled independently, the trials do not establish a causal directional relationship. They do show that apparent contact changed substantially during otherwise similar manual sweeps.

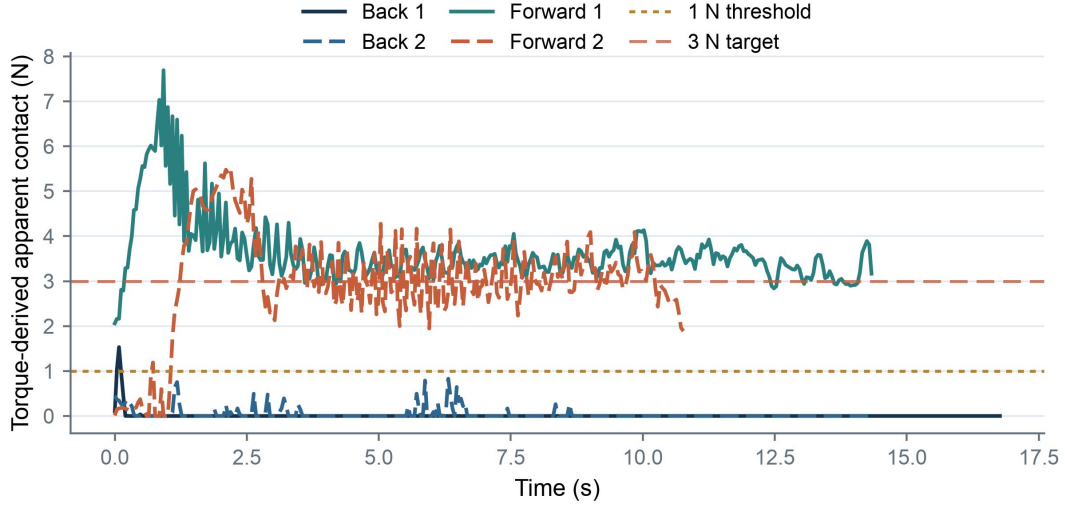


Figure 6.4. Torque-derived apparent contact estimate during four repeated manual-guidance trials. Differences among the trials illustrate interaction- and configuration-dependent coupling in the FR3 wrench estimate.

## 6.4 Combined Guidance and Contact

In the two preliminary Pose 4 trials on the soft, skin-like surface, the torque-derived estimate averaged  $2.54 \pm 0.08$  N. RMSE relative to the nominal internal 3 N target was  $1.02 \pm 0.14$  N, and the mean within-trial standard deviation was  $0.90 \pm 0.20$  N. Approximately 5.4% of samples were at or below the 1 N contact threshold. Figure 6.5 presents one representative trial.

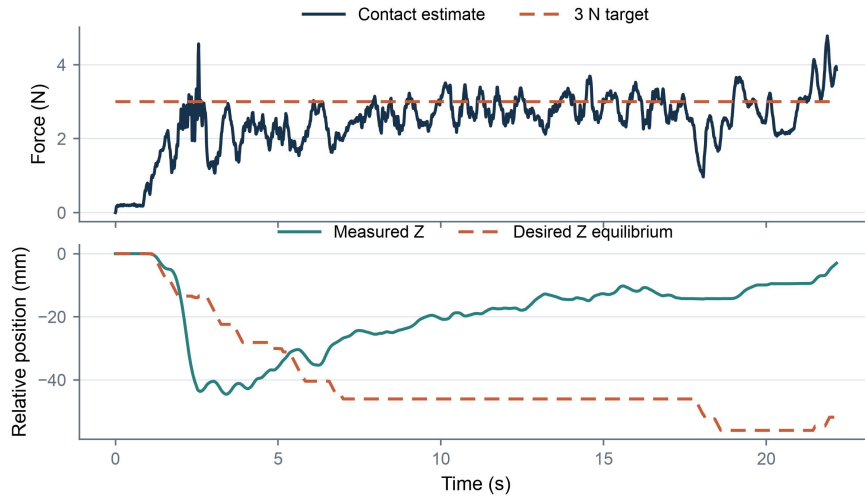


Figure 6.5. Torque-derived contact estimate and relative Z response during a representative Pose 4 soft-surface contact trial. No independent force measurement or ultrasound acquisition was available.

Across the two trials, the maximum absolute desired Z-equilibrium offset was  $84 \pm 39$  mm and the measured Z range was  $38 \pm 9$  mm. These values describe the scale of motion produced

by the combined compliance and equilibrium adaptation; they are not force or safety acceptance limits.

Pose 4 was exploratory. Only two repetitions were recorded, the soft surface had no independent instrumentation, and neither surface deformation nor ultrasound image quality was measured. The trials show that manual guidance and contact-related control could operate simultaneously, but they also show the need for direct sensing and conservative, application-specific travel limits.

## 6.5 Discussion and Limitations

The configured impedance allowed the FR3 to be moved by hand while Cartesian position and selected controller variables were recorded. Pose 3 provides the quantitative evidence for this translation. Lower rotational stiffness and reduced joint-6 task torque also seemed to make orientation changes easier, but orientation was not logged and no controlled angular comparison was performed. That observation remains qualitative. Future ultrasound registration will require full six-degree-of-freedom pose logging.

No additional force-signal filter was used in the v5.1 controller or in the reported analysis. Processing consisted of subtracting one startup bias sample and rectifying the corrected  $Z$  component. Random noise was not the main problem because operator input, surface contact, robot configuration, and modelling error all affected the same FR3 wrench estimate. A separate axial sensor in the probe load path is needed before true contact-force accuracy can be measured.

The experiments did not include ultrasound acquisition, image-pose synchronization, spatial calibration, or volume reconstruction. These imaging tasks should begin after the direct force sensor has been integrated and calibrated.

## **6.6 Chapter Summary**

The trials showed that the impedance permitted manual displacement, but the torque-derived estimate could not distinguish surface contact from operator input. The stationary results concern only internal-estimate regulation; direct axial sensing is required for further validation.

## 7. Conclusions, Recommendations and Future Work

### 7.1 Conclusions

This project developed a low-level FR3 interaction controller as an early step towards autonomous robotic 3D ultrasound. The robot could be translated by hand, remained manually adjustable in Z while the controller updated the Z equilibrium, and allowed low-resistance orientation changes. The program also provided controlled stopping and timestamped logging. Together, these functions form the interaction layer needed before a probe can be integrated.

The controller kept the torque-derived contact estimate near its target during relatively undisturbed stationary contact. During manual movement, the same estimate changed with both operator input and surface contact. The experiments therefore did not demonstrate accurate regulation of the true probe-skin force. The main limitation was the measurement architecture, not the basic operation of the impedance controller.

The completed system is a human-guided interaction and logging prototype, not an autonomous or clinically validated ultrasound platform. The next step is to integrate the axial force sensor and transducer holder, calibrate the assembled load path, and repeat the contact tests on a fixed phantom while the robot is moving. Ultrasound synchronization and spatial calibration can then be added before work on surface following, autonomous paths, or image-quality feedback.

### 7.2 Recommendations

The immediate recommendation is to complete the external-sensor load path using the Honeywell FSS1500NSB, PhidgetBridge, and transducer-holder prototype. The assembled system should be calibrated with known loads in its final mounting orientation. Sensor zero, gain, hysteresis, drift, saturation, sample timing, and the transformation between the sensor normal and FR3 coordinates should be documented before closed-loop use. The selected force sensor is shown in Figure 7.1.





*Figure 7.1. Honeywell FSS1500NSB force sensor selected for future direct contact-force measurement [10]; it was not used in the Chapter 6 experiments.*

The protocols for Pose 1 through Pose 4 should then be repeated while recording both the direct sensor and the FR3 estimate. Pose 3 provides the most important validation: the direct normal-force signal should remain near zero during no-contact XY guidance even when the torque-derived estimate changes. The two signals can then be compared using bias, RMSE, correlation, delay, and cross-axis sensitivity.

Future probe-contact tests should set and document the Z-equilibrium speed and travel limit according to holder geometry, surface location, and allowable tissue displacement. Contact-loss, overload, communication-failure, and user-stop responses should be tested at low speed on a non-human phantom before any human interaction is repeated.

### 7.3 Future Work

Once the direct force signal is validated, it can regulate normal contact by adapting the Z equilibrium while preserving compliant manual motion in both axial and lateral directions. The FR3 torque estimate or a separate interaction channel may continue to support operator guidance. If finer speed control is required, proportional admittance could replace the fixed-magnitude XY command. Orientation assistance could later progress from low stiffness to surface-normal following after surface geometry can be estimated reliably.

The imaging stage requires completion of the transducer holder, rigid flange-to-image calibration, and synchronized recording of B-mode frames and FR3 poses. Only then can tracked 2D frames be compounded into a 3D volume and evaluated for coverage, geometric consistency, missing frames, and sensitivity to contact-force variation. Initial imaging experiments should use a fixed ultrasound phantom.

Longer-term work may examine planned scan corridors, automatic surface-normal estimation, image-quality feedback, and comparison with freehand acquisition. These functions should follow repeatable force measurement, mechanical calibration, image synchronization, and safety validation.

## 7.4 Project Limitations

The principal limitation was the absence of an independent end-effector force sensor during development and final logging. The study therefore measured the FR3 torque-derived estimate rather than true probe-surface force. The dataset included only two or three repetitions for Pose 1, Pose 3, and Pose 4, four selected steady-state repetitions for the primary Pose 2 analysis, and four directional-guidance trials. The steady-state windows were selected retrospectively because the logs lacked manual event markers; excluded Pose 2 recordings contained major transients, repeated contact loss, or unintended guidance activation. Pose 2 and Pose 4 used a soft, skin-like surface without independent instrumentation. The analysis is therefore descriptive and supports no claims about clinical safety, tissue loading, or diagnostic image quality.

No ultrasound frames were acquired, and no 3D reconstruction was performed. Earlier controller versions were not logged systematically, so comparisons among versions remain partly qualitative. The conclusions are limited to one FR3, one v5.1 implementation, the preliminary 30 July dataset, and the documented laboratory conditions.

## 7.5 Self-Assessment

The project began with simulation, ROS 2, MoveIt, path planning, and 3D reconstruction all under consideration. As the work progressed, physical interaction became a more immediate problem than trajectory planning. MoveIt was useful for learning pose and path execution, but it was not used for the final real-time controller. Other controller frameworks added dependency, real-time, and hardware-interface difficulties. Starting from the official libfranka Cartesian impedance example made the system simple enough to test and modify directly on the robot.

The greatest difficulty was not a single equation or software defect, but the integration of robot hardware, real-time control, medical-imaging requirements, and imperfect sensing. Each stage changed the interpretation of the previous one. The controller evolved from planned motion

toward human guidance, while the force problem developed from a tuning question into recognition that the available estimate could not distinguish operator input from surface contact.

The no-contact trials on 30 July showed that a stable-looking robot signal is not necessarily a direct measurement of the quantity being studied. If I restarted the project, I would introduce systematic logging, predefined test conditions, and direct force-sensor planning earlier. This would make it easier to compare controller versions using data instead of relying mainly on how the robot felt during operation.

The final controller met the basic interaction objective, but its Z-force estimate was not adequate for force validation. The report records what was attempted, what changed during development, what the prototype could do, and where the data showed its limits. It does not present the system as finished.

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## Appendix A. Experimental Data Manifest

The analysis used 19 timestamped v5.1 CSV files. Raw samples were retained; no filtering was applied to the reported force-estimate results. File names and roles are listed below.

### A.1 Log Files

| Condition   | Files                                                       | Use                                                              |
|-------------|-------------------------------------------------------------|------------------------------------------------------------------|
| Pose 1      | 134806, 134825, 134846                                      | No-contact baseline                                              |
| Pose 2      | 135505, 135614, 135933, 150749,<br>151143, 151247, 151323   | Stationary-contact and transient<br>assessment                   |
| Pose 3      | 140308, 140336, 140402                                      | Manual XY movement without<br>intended surface contact           |
| Pose 4      | 140545, 140700                                              | Combined guidance and soft-surface<br>contact                    |
| Directional | 141256 back, 141338 back, 141317<br>forward, 153653 forward | Interaction-coupling comparison across<br>repeated manual sweeps |

The primary Pose 2 steady-state windows were 10.000-17.314 s (135505), 12.000-33.193 s (151143), 4.000-30.494 s (151247), and 2.000-26.194 s (151323). They were selected retrospectively after contact establishment using the qualitative rule in Section 5.8. The excluded trials were 135614 (large initial Z transient), 135933 (substantial XY activation during the nominally stationary condition), and 150749 (repeated contact loss). The selected intervals are not equivalent to a prospectively defined acceptance test.

### A.2 Analysis Definitions

- XY command active:  $\sqrt{v_{des,x}^2 + v_{des,y}^2} > 10^{-5} \text{m/s}$ .
- Rectified contact estimate:  $\hat{F}_{contact} = \max(-F_{z,corr}, 0)$ .
- Apparent contact threshold: 1 N in the rectified estimate.
- Target-deviation RMSE: root-mean-square difference between the rectified estimate and the nominal internal 3 N target; it is not error relative to an independent force sensor.
- Trial summaries were calculated per trial and then summarized across trial-level values so unequal log durations did not create independent repetitions.

## Appendix B. Controller Algorithm and Reproducibility Notes

The complete controller is supplied electronically as `impedance_control_v5.1.c`. Despite the historical `.c` extension, the file contains C++ and was compiled as part of the libfranka C++ examples. Renaming it to `impedance_control_v5.1.cpp` is recommended for future builds. The implemented cycle can be summarized as follows:

1. Read one initial robot state and store its external-wrench components as the bias.
2. At each torque-control cycle, subtract the bias and rectify negative corrected Z force into a non-negative contact estimate.
3. Generate a fixed-magnitude desired XY velocity when the horizontal wrench norm exceeds 1 N.
4. Use Cartesian velocity damping and saturate XY task force at 10 N.
5. Move the desired Z equilibrium at 0.02 m/s when the rectified estimate is outside the  $3 \pm 1$  N deadband and above the 1 N contact threshold.
6. Limit Z task force to 10 N and rotational moment to 3 N·m.
7. Map the task wrench through the Jacobian transpose, add Coriolis compensation, and retain 30% of Cartesian task torque at joint 6.
8. Buffer selected variables at approximately 100 Hz and write the timestamped CSV after control exits.